

Auteur: Karanjot Singh
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$$\int \frac{dx}{\sqrt{-x^2 + 4x - 3}}$$
$$\sqrt{-x^2 + 4x - 3} = \sqrt{-(x-2)^2 + 1}$$

Stel $u = x - 2$

$$\Rightarrow du = dx$$

$$= \int \frac{dx}{\sqrt{-(x-2)^2 + 1}}$$
$$= \int \frac{du}{\sqrt{1-u^2}}$$
$$\int \frac{du}{\sqrt{1-u^2}} = \text{Bgsin}(u) + C$$

$$\text{Bgsin}(u) = \text{Bgsin}(x-2)$$

$$\Rightarrow \int \frac{dx}{\sqrt{-x^2 + 4x - 3}} = \text{Bgsin}(x-2) + C$$

References